

A Cost-Effective Predictive HR Analytics Framework for Sri Lankan SMEs Using Cloud-Native Serverless AI

Student Name	Theekshana Gimhan
Student ID	15002
Degree Programme	BSc (Hons) Management Information Systems
Module	COM 4901 – Final Year Individual Research Project
University	Kaatsu International University
Faculty	Faculty of Computer Science and Engineering
Supervisor	Ms. Thanuja Irugalbandara
Date	March 2026

*Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Science Honours in Management Information Systems,
Kaatsu International University, March 2026.*

Title

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Introduction / Background

Small and Medium Enterprises (SMEs) form the backbone of Sri Lanka's economy, contributing approximately 52% of the national Gross Domestic Product (GDP) and providing employment to 45% of the workforce [1]. Despite their economic significance, the majority of Sri Lankan SMEs continue to rely on manual, paper-based processes or basic spreadsheet-driven approaches for managing their human resources [2]. This reactive management model means that critical decisions — such as identifying employees at risk of resignation — are made only after the damage has occurred, leading to disruption, increased recruitment costs, and loss of institutional knowledge.

The global Human Resource Technology (HR Tech) market has responded to this challenge with sophisticated predictive analytics platforms. Enterprise solutions such as SAP SuccessFactors, Workday, and Oracle HCM Cloud now offer AI-powered attrition prediction, workforce planning, and sentiment analysis. However, Sri Lanka's digital economy landscape, characterised by rapidly growing technology adoption yet constrained resources among SMEs, makes these solutions particularly difficult to access [3]. These platforms typically operate on a per-employee-per-month (PEPM) pricing model ranging from USD 8–15 per employee, translating to approximately LKR 240,000–450,000 annually for a 50-employee SME [4]. This cost structure renders enterprise HR analytics financially inaccessible for the very businesses that would benefit most from data-driven workforce decisions.

Recent advancements in cloud-native serverless computing, particularly Google Cloud's Vertex AI platform and managed Cloud Functions, have fundamentally altered the economics of deploying machine learning models [5]. Serverless architectures eliminate the need for always-on infrastructure, enabling a “pay-per-prediction” model that significantly reduces operational costs compared to traditional virtual machine deployments [6]. This paradigm shift presents a unique opportunity to deliver enterprise-grade predictive analytics to resource-constrained SMEs at a fraction of the traditional cost.

This project proposes the design and development of an AI-powered predictive analytics module integrated into *NexusHR*, an existing cloud-native Human Resource Management System (HRMS) built on modern web technologies (React, Node.js, PostgreSQL) and deployed on Google Cloud Run. The module will leverage serverless AI services to forecast employee attrition risk, providing SME managers with actionable insights through an intuitive Insights Dashboard — all within a monthly operational budget of LKR 10,000.

This research directly aligns with Sri Lanka’s National Artificial Intelligence Strategy (2023–2028) and Digital Economy Strategy 2030, which emphasise the democratisation of AI technologies for inclusive economic growth [7].

Problem Statement

Employee turnover poses a significant financial and operational burden on Sri Lankan SMEs. Research indicates that the cost of replacing a departing employee ranges from 50% to 200% of their annual salary when accounting for recruitment, onboarding, training, and productivity losses during the transition period [8]. For SMEs operating on thin margins, unplanned attrition can be existentially threatening. In the current Sri Lankan economic context, the emigration of skilled professionals — commonly referred to as the “Brain Drain” — has intensified, with over 300,000 Sri Lankans departing for overseas employment annually [17]. This trend makes the retention of remaining domestic staff a critical survival factor for SMEs, further underscoring the urgency of proactive, data-driven retention strategies.

Despite the demonstrated effectiveness of machine learning in predicting employee attrition — with studies achieving accuracy rates of 85–95% using algorithms such as Random Forest, XGBoost, and Gradient Boosting [9, 10] — the practical adoption of such models in SME environments remains critically low. Three primary barriers contribute to this gap:

1. **Cost Barrier:** Enterprise predictive HR platforms (Workday, SAP) operate on per-employee-per-month pricing of approximately USD 8–15 (LKR ~2,400–4,500/employee/month), resulting in annual costs exceeding LKR 240,000 for a 50-employee firm [4]. In contrast, this research targets a total monthly operational cost of LKR 10,000 — equivalent to just LKR 200 per employee per month (approximately USD 0.60) — a reduction of over 90% compared to enterprise alternatives.
2. **Infrastructure Barrier:** On-premise AI deployments require dedicated GPU hardware, specialised DevOps personnel, and ongoing maintenance — resources unavailable to most SMEs [11].
3. **Integration Barrier:** Standalone AI tools operate in isolation from existing HR workflows, requiring manual data transfers and producing insights disconnected from operational context [12].

There is a clear research gap in the development of a **cost-optimised, cloud-native predictive analytics framework** that is (a) financially accessible to SMEs with monthly costs under LKR 10,000, (b) architecturally integrated into an existing HRMS to eliminate data silos, and (c) contextualised for the Sri Lankan employment landscape, including statutory regulations (EPF, ETF) and local workforce dynamics.

Research Questions

Based on the identified research gap, this study seeks to answer the following questions:

- **RQ1:** What are the most significant behavioural predictors of employee attrition derivable from standard HR operational data (attendance, leave, payroll) within Sri Lankan SMEs?
- **RQ2:** To what extent can a serverless cloud architecture reduce the operational cost of deploying HR predictive analytics compared to traditional persistent infrastructure, and can it remain within a monthly budget of LKR 10,000?
- **RQ3:** Can a hybrid dataset approach — combining locally augmented synthetic data with global benchmark datasets — achieve an F1-Score of $\geq 80\%$ for employee attrition classification in low-volume SME data environments?

Research Aim and Objectives

Aim: To design, develop, and evaluate a cost-effective, serverless AI-driven predictive analytics module integrated into an existing cloud-native HRMS to forecast employee attrition risk for Sri Lankan SMEs.

Objectives:

- O1. Feature Identification:** To identify and extract key behavioural predictors of employee attrition from HR operational data — including attendance patterns, leave frequency and type, salary positioning, tenure, and payroll anomalies — within the Sri Lankan SME context, ensuring compliance with the Sri Lanka Personal Data Protection Act No. 9 of 2022 [15].
- O2. Architecture Design:** To design a serverless cloud architecture utilising Google Vertex AI and Cloud Functions that maintains monthly operational costs low for a typical 20–50 employee organisation, establishing a verifiable cost-efficiency baseline against traditional always-on VM deployments.
- O3. Model Development:** To develop and train a machine learning classification model using a hybrid dataset approach (synthetic data augmented with the IBM HR Analytics dataset) to achieve a Recall score of $\geq 80\%$ in identifying high-risk attrition cases, thereby minimising false negatives. The 80% threshold is grounded in the asymmetric cost structure of HR prediction: a false negative (failing to identify an employee who subsequently resigns) incurs replacement costs of 50–200% of annual salary [8], whereas a false positive (incorrectly flagging a stable employee) incurs only the minimal cost of a proactive retention conversation. Should the target recall not be achieved with the initial dataset, the research will pivot to identifying the critical

data volume threshold required for reliable AI-driven attrition prediction in SME environments — a finding that is itself a significant contribution to the field.

O4. System Integration: To implement a fully integrated Insights Dashboard within the NexusHR frontend that visualises employee risk scores, feature-level explanations using SHAP (SHapley Additive exPlanations) values, and trend analysis to support managerial decision-making.

O5. Evaluation: To evaluate the framework’s effectiveness through (a) predictive accuracy metrics (Recall, F1-Score, AUPRC), (b) a comparative cost analysis across three deployment architectures (serverless, container-based, and hybrid), and (c) a usability evaluation of the Insights Dashboard using the System Usability Scale (SUS) [16].

Scope and Deliverables

In-Scope:

- Employee attrition prediction using binary classification (At-Risk / Not At-Risk)
- Serverless batch prediction pipeline (Cloud Functions + Cloud Scheduler) for nightly predictions
- Insights Dashboard with risk scores, SHAP-based feature explanations, and historical trend visualisations
- Cost benchmarking across three deployment architectures
- Synthetic data generation pipeline for model training
- Privacy-preserving data extraction (PII anonymisation, tenant isolation)

Out-of-Scope:

- Real-time sentiment analysis (e.g., email or chat monitoring)
- Mobile application development
- Modifications to existing payroll, leave, or attendance processing logic in the base HRMS
- Multi-language natural language processing features
- Performance prediction or succession planning modules

Deliverables:

1. A functional AI-enhanced HRMS prototype deployed on Google Cloud Platform
2. A trained and evaluated ML classification model with documented metrics

3. A serverless batch prediction pipeline with automated scheduling
4. A comparative architecture cost analysis report with real GCP billing data
5. A comprehensive research dissertation

Literature Summary

The following is a brief overview of the key literature informing this research, organised into three thematic areas.

Machine Learning for Employee Attrition Prediction

The application of machine learning to employee attrition prediction has been extensively studied. Punnoose and Ajit (2016) demonstrated that ensemble methods — particularly Random Forest and Gradient Boosting — consistently outperform traditional logistic regression for turnover classification, achieving accuracy rates exceeding 85% [10]. Sarker (2021) provided a comprehensive survey of ML algorithms and their real-world applications, noting that tree-based ensemble methods are particularly effective for tabular HR data due to their ability to capture non-linear feature interactions [9]. The IBM HR Analytics dataset has become a widely used benchmark, featuring 35 features across 1,470 employee records with labelled attrition outcomes [14]. However, a significant limitation of existing studies is their reliance on datasets from Western corporate environments, with limited validation in developing economy SME contexts.

Serverless and Cloud-Native AI Economics

The economics of serverless computing for AI workloads have been examined by Kodakandla (2021), who demonstrated through a comparative study that serverless architectures offer significant cost and scalability advantages for cloud-native applications — particularly for intermittent workloads, precisely the usage pattern of nightly batch HR predictions [6]. Google Cloud’s Vertex AI platform enables a “pay-per-prediction” model that eliminates idle compute costs, making enterprise-grade ML accessible to smaller organisations [5]. Ribeiro et al. (2015) introduced the Machine Learning as a Service (MLaaS) paradigm, arguing that cloud-based ML reduces the infrastructure expertise barrier for non-technical organisations [11]. However, no existing study has quantified the cost of deploying predictive HR analytics specifically within an SME budget constraint (under LKR 10,000/month).

The Digital Divide in Sri Lankan SMEs

Despite contributing 52% of GDP, Sri Lankan SMEs face significant barriers to digital transformation, including limited financial resources, shortage of skilled personnel, and inadequate digital infrastructure [1, 2]. The 2023 Brain Drain phenomenon has further strained domestic human capital, making employee retention a strategic imperative [17].

Enterprise HR technology solutions remain financially inaccessible — industry pricing analyses indicate that cloud HCM suite pricing ranges from USD 8–15 per employee per month [4]. Angrave et al. (2016) identified that even when organisations invest in HR analytics, the disconnection between analytics tools and operational HR systems creates data silos that undermine practical utility [12]. This research addresses this gap by integrating predictive analytics directly into an existing operational HRMS.

Proposed Methodology

This project will follow a **Design Science Research (DSR)** methodology, which is specifically suited for development-oriented research where the creation and evaluation of an IT artifact constitutes the primary research contribution [13]. The DSR framework comprises the following phases:

Phase 1: Problem Identification and Literature Review (Month 1)

- Comprehensive literature review on employee attrition prediction, serverless ML architectures, and cost-optimised AI deployment
- Analysis of existing predictive HR analytics solutions and their cost structures
- Formalisation of the research gap and evaluation criteria

Phase 2: Data Preparation and Feature Engineering (Month 1–2)

- Extraction of anonymised HR operational data from the NexusHR PostgreSQL database in compliance with the Sri Lanka Personal Data Protection Act No. 9 of 2022 [15]
- Development of a statistical data augmentation pipeline that generates approximately 500 synthetic records from the existing 20–50 real employee records, modelling the statistical distributions of the original Sri Lankan workforce data
- Supplementation with the IBM HR Analytics dataset [14] to increase sample diversity and provide labelled attrition outcomes
- Feature engineering: absenteeism rate (3-month and 6-month rolling), leave pattern variability, overtime trends, salary percentile within department, and tenure buckets
- Application of SMOTE (Synthetic Minority Over-sampling Technique) to address class imbalance
- Privacy preservation through PII stripping and multi-tenant data isolation

Phase 3: Model Development and Training (Month 2–3)

- Baseline model training using Python (Scikit-learn): Logistic Regression, Random Forest, XGBoost, Gradient Boosting
- Hyperparameter optimisation using cross-validation
- Deployment of the best-performing model to Google Vertex AI for cloud-based inference
- Comparison of AutoML Tabular vs. custom-trained model performance
- Integration of SHAP (SHapley Additive exPlanations) for model interpretability

Phase 4: Architecture Design and Cloud Integration (Month 3–4)

- Design and implementation of the primary serverless batch prediction pipeline (Architecture A):
 - Google Cloud Scheduler triggers nightly Cloud Function
 - Cloud Function extracts features from PostgreSQL, invokes Vertex AI prediction endpoint
 - Results stored back in NexusHR database (`PredictionResult` and `PredictionRun` tables)
- Comparative cost analysis across three deployment architectures:
 - **Architecture A:** Serverless (Cloud Functions + Vertex AI) — *primary build*, zero idle costs
 - **Architecture B:** Container-based (Cloud Run with embedded model) — cost-projected using GCP Pricing Calculator
 - **Architecture C:** Hybrid (Cloud Run for orchestration, Vertex AI for inference) — cost-projected using the same methodology
- *Note:* The system prototype is developed under the working title *Simpala HR*, while *NexusHR* serves as the academic designation for the research framework

Phase 5: Frontend Integration and Dashboard Development (Month 4)

- Development of the Insights Dashboard within the NexusHR React frontend:
 - Employee Risk Score cards with colour-coded severity indicators
 - SHAP-based feature importance visualisations
 - Historical trend charts showing risk score progression over time
 - Department-level risk heatmap for aggregate workforce analysis
- Role-based access control integration (Owner/Admin only)

Phase 6: Evaluation and Dissertation (Month 4–5)

- **Predictive Accuracy:** Confusion matrices, ROC curves, Precision-Recall curves, F1-Score, and AUPRC using Stratified k -Fold Cross-Validation ($k=5$)
- **Cost-Efficiency:** Real GCP billing data analysis comparing three deployment architectures against the LKR 10,000 monthly budget target using Google Cloud Billing API export logs
- **Usability:** System Usability Scale (SUS) questionnaire [16] administered to 3–5 SME stakeholders to obtain a standardised usability score (0–100), supplemented by structured qualitative feedback
- Compilation of results and completion of the research dissertation

Tools and Resources

Category	Tools
Base HRMS	NexusHR — React 19, Node.js/Express 5, Prisma 6 ORM, PostgreSQL
Cloud Platform	Google Cloud Platform (Cloud Run, Cloud Functions, Cloud Scheduler, Vertex AI, BigQuery)
ML/AI	Python 3.x, Scikit-learn, XGBoost, Pandas, NumPy, SHAP, Matplotlib
Data	IBM HR Analytics Employee Attrition Dataset [14], Synthetic data via statistical augmentation
Development	TypeScript, Vite, MUI (Material UI), Git/GitHub
CI/CD	GitHub Actions (existing pipeline), Docker, Google Artifact Registry
Evaluation	Google Cloud Billing API, Confusion Matrix analysis, AUPRC metrics, SUS Questionnaire

Table 1: Tools and resources used in the project

Preliminary Timeline

Ethical Considerations

This project involves processing employee HR data, which raises the following ethical considerations:

- **Legal Compliance:** All data processing activities will be conducted in accordance with the **Sri Lanka Personal Data Protection Act No. 9 of 2022** [15], which governs the collection, processing, and storage of personal data. The project will ensure that data processing is limited to the stated research purpose (attrition prediction),

Month	Phase	Key Activities
Month 1 (Mar 2026)	Literature Review & Data Gathering	Finalise literature review; analyse IBM dataset; begin feature engineering; ethical considerations
Month 2 (Apr 2026)	Data Preprocessing & Model Training	Complete feature engineering; train baseline ML models; evaluate initial performance metrics
Month 3 (May 2026)	Cloud Architecture & Deployment	Deploy to Vertex AI; implement serverless pipeline; begin cost benchmarking
Month 4 (Jun 2026)	Integration & Dashboard	Build Insights Dashboard; integrate prediction results into NexusHR; Interim Report (29 June)
Month 5 (Jul–Aug 2026)	Evaluation & Dissertation	Complete cost analysis; usability assessment; write final dissertation; Final Report (31 August)

Table 2: Preliminary project timeline

that appropriate anonymisation measures are applied, and that data subject rights are respected.

- **Data Privacy:** All employee data extracted from the HRMS will be anonymised by removing Personally Identifiable Information (PII) — including names, NIC numbers, email addresses, and bank details — before any ML processing. Only aggregated behavioural features (attendance rates, leave frequency, tenure metrics) will be used for model training.
- **Informed Consent:** As the system uses aggregated behavioural patterns rather than individual surveillance, the ethical implications are minimised. However, clear disclosure will be made to stakeholders about the nature of the predictive module.
- **Bias Mitigation:** The training data will be audited for demographic biases (gender, department) to ensure the model does not unfairly flag specific employee groups.

- **Multi-Tenant Isolation:** The system’s existing multi-tenant architecture ensures that data from different companies is never mixed during analysis.

No ethical approval is anticipated to be required as the project uses anonymised data from an existing system and synthetic datasets. This will be confirmed with the supervisor.

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